PARTIAL CHECKPOINT | PLAN 496

Two model pairs, one engineering gate

Snapshot only: plan 496. Subsequent source preparation or live operations are outside this report.

Quad goal 3 seeds 11 and 23 each completed fresh BC50k + QV50k, with independent terminal audits and ROOT reconciliation. Original QGF/FK public-reference engineering passed 60 cells. All Quad guided native success rates remain UNRUN.

Completed training	Engineering result	Native performance
4 / 18 phases; 2 / 9 pairs	QGF 12 + FK 48: 60 PASS	QGF / 8DF / IPG / FK / BC: UNRUN

All nine goal x training-seed lineages retained

Goal	Seed 11	Seed23	Seed37
3	BC50k + QV50k complete	BC50k + QV50k complete	BC UNRUN / QV UNRUN
4	BC UNRUN / QV UNRUN	BC UNRUN / QV UNRUN	BC UNRUN / QV UNRUN
5	BC UNRUN / QV UNRUN	BC UNRUN / QV UNRUN	BC UNRUN / QV UNRUN

Three training-seed values (11,23,37), not nine independent seeds. Remaining: 14 phases = 7 BC + 7 QV = 700,000 effective updates. Current effective updates: 200,000; physical returns: 200,723, retaining the old 723 repeated QV returns. BC37 was source preparation only at cutoff.

Local pilot context - not paper-scale replication

Dimension / setting	This pilot	Paper context
Observation / native action / chunk	55 / 5 / H5 flat25	Single-task Cube setting
Data / width / updates	One official TRAIN000 shard; local 1M / 128x128 / 50k each	100M / 1024x4 / 500k
Flow / temperature / augmentation	F8 / T1 / unaugmented	F10 context; not matched here

P1 found 996,000 full-H5 eligible starts per goal and zero strict joint-goal TRAIN rows. That describes this dataset; it is not a competence, impossibility, Q-truth, or hidden BC success gate. [S00,S02,S17]

Training history and saved gates

Updates, API calls, dataset views and native environment controls are different units.

Completed goal 3 training phases

Seed / phase	Effective result	Outer parent seconds
11 BC	50,000 fresh; 51 durable pairs	911.232648
11 QV original	15,723 returned; durable15,000; bounded failure	292.626877
11 QV continuation	35,000 new, indices15000..49999; effective50k	603.704228
23 BC	50,000 fresh; 51 durable pairs	919.656290
23 QV	50,000 fresh; 51 durable pairs	991.807003

The original seed 11 QV attempt remains FAILED/BOUNDED_PARTIAL. Its 723 returned but uncheckpointed updates were discarded and repeated by the exact continuation. Trigger operands were not persisted, so the stop cause remains UNKNOWN. The continuation passed 70,446 saved predicates and restored the final 50,000 fullstate from disk with semantic and canonical-byte checks.

Each QV phase freezes the actual matching BC actor FULL TrainState, Adam state and agent RNG. Seed 11 continuation records one fresh matching-template proof plus 74 frozen-parent proofs. Final fullstate payloads are references only in this report; no weights were read or decoded.

Quad sampler readiness at this cutoff

Gate / family	Saved result	What it supports
QGF 12 / FK 48	60 PASS; 180 sampler +3 helper B20; 24 inventories	Original public-reference engineering at this 55D frozen model
Engineering artifacts	60NPZ / 3024 header descriptors / 4,491,360B; 3 full proofs /2 canonical	Saved integrity and declared public output agreement
DF128	120 PASS / 8 latent FAIL; max 1.220703125e-4 vs 2e-6; actions PASS	DF gate failed; f64 UNRUN, no DF40/native release
IPG	Unqualified	No solver/domain/tolerance bridge or hidden qualification

QGF/FK outer71.3557048s was counted once after audit. TRAIN16 +4 padding are engineering rows, not20 independent episodes. These passes do not repair DF, establish hidden-ancestor equality, true Q, native utility, or algorithm superiority. [S08-S16]

Primary Task2: own-best confirmation

PointMaze Task 2 results are a separate campaign. They are not Quad native measurements.

Fresh40 successes from each frozen selected setting

Mode / critic	lambda	Seed 11	Seed23	Seed37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	40/40
Stochastic / averaged Q	1	35/40	40/40	40/40
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	UNQUALIFIED	UNQUALIFIED	UNQUALIFIED

Thirty of the required 33 guided selections are frozen: DF24 + QGF3 + FK3. IPG3 remain unqualified. The explicit KEEP_ALL33 barrier is unchanged. New main100 heldout episodes:0; the planned540 matched-curve points x100 remain pending.

What these numbers do and do not compare

Every cell uses its own selected configuration and fresh40 confirmation. Settings, cohorts and cumulative guidance differ. This is not a matched-kappa averaging experiment, a heldout estimate, or evidence that tuned DFlow dominates QGF/FK. Equal numeric kappa across methods is not equal accumulated guidance.

The complete 24 selected DF tuples are on page4 and in the CSV sidecar. All72 literal fresh40 DF alternatives remain intact in DF72_LITERAL_FRESH40_ALTERNATIVES.csv. No fourth candidate, new retuning, or exhaustive Cartesian result is invented.

Primary closed native VAL/CONF: 30,360 episodes and 16,185,534 action rows. Separate Triple BC20:20 episodes /20,000 external controls, not added to that primary total. Its observed0/20 is a BC result, not QGF failure. [S02,S04-S07; selected CSVs]

All 24 frozen DFlow settings

Each row is an own-best fresh40 choice; lambda is the prior coefficient. M8 / T1 / N1 throughout.

Seed	Mode/Q	L	kappa	eta	I	sigma	K	CONF
11	reg/raw	0	0.1	0.1	256	0	4	35/40
11	reg/raw	1	40	0.03	256	0	4	28/40
11	sto/raw	0	5	0.1	128	0	4	40/40
11	sto/raw	1	5	0.1	128	0	4	37/40
11	reg/avg	0	1	0.1	256	0.003	16	38/40
11	reg/avg	1	64	0.03	256	0.5	4	25/40
11	sto/avg	0	5	0.1	128	0.003	16	40/40
11	sto/avg	1	5	0.03	128	0.003	16	35/40
23	reg/raw	0	5	0.1	256	0	4	40/40
23	reg/raw	1	64	0.03	256	0	4	40/40
23	sto/raw	0	80	0.03	256	0	4	40/40
23	sto/raw	1	80	0.01	128	0	4	40/40
23	reg/avg	0	16	0.1	256	0.3	4	40/40
23	reg/avg	1	80	0.03	128	0.5	16	40/40
23	sto/avg	0	64	0.1	128	0.5	4	40/40
23	sto/avg	1	80	0.1	256	0.01	4	40/40
37	reg/raw	0	0.1	0.1	256	0	4	37/40
37	reg/raw	1	5	0.1	128	0	4	40/40
37	sto/raw	0	5	0.1	128	0	4	39/40
37	sto/raw	1	5	0.1	128	0	4	40/40
37	reg/avg	0	0.1	0.1	256	0.003	4	37/40
37	reg/avg	1	5	0.03	256	0.003	4	40/40
37	sto/avg	0	3	0.1	256	0.003	4	40/40
37	sto/avg	1	5	0.03	128	0.003	4	40/40

All eight variants retain the identity/full-F8 path: regular or stochastic x raw or averaged Q x lambda 0 or 1. Raw-Q rows carry sigma 0 / K4 metadata; no averaging benefit is inferred from that bookkeeping.

DF hierarchy: 20 kappa values from 0 to 80; eta {.001,.003,.005,.01,.03,.1}; I {1,2,4,8,16,32,64,128,256}; averaged-Q sigma {.003,.01,.03,.06,.1,.2,.3,.5}; K {4,16}. The adopted hierarchy is 9,240 settings/model (27,720 saved TRAIN scores), not 73,440 Cartesian settings.

Config IDs, selection hashes, full-agent/actor hashes and all original tuple fields are retained in DF24_SELECTED_TUPLES.csv; exact 72 alternative ordering and outcomes are retained separately. Frozen aggregate19d8af37... [S04; CSV provenance manifest]

Q averaging, coefficients and controls

A sampler engineering pass does not measure return or prove that a learned critic is correct.

What the local average computes

Qavg(s,u) = (1/K) sum_k Q(s, u + sigma * epsilon_k), using a fixed nearby-noise bank. The current bank is antithetic. The unperturbed center is not explicitly added as a separate sample. Averaging queries Q at nearby inputs; it does not require new transitions or a value-network query.

Local smoothing can soften a false narrow peak or a real narrow optimum. It does not automatically fix broad critic scale errors, out-of-distribution extrapolation, or Q truth. No causal averaging gain, safe repair, or DFlow advantage has been established by these selected-setting outcomes.

Selected QGF/FK settings from separate native campaigns

Method	Seed	kappa	M	N	noise	CONF
T2 QGF	11	16	64	1	0	40/40
T2 QGF	23	320	8	1	0	40/40
T2 QGF	37	16	16	1	0	40/40
T2 FK	11	80	64	32	0.02	40/40
T2 FK	23	80	32	32	0.02	40/40
T2 FK	37	80	16	32	0.02	40/40
T5 QGF	11	96	16	1	0	40/40
T5 QGF	23	2	64	1	0	36/40
T5 QGF	37	80	32	1	0	40/40

Task 5 QGF 40/36/40 includes four seed 23 failures; it is not generic QGF collapse. These campaigns and the Quad 55D pilot have different tasks, models and settings.

kappa scales guidance; eta is DF outer-update size; I is DF source iteration count; M is integration steps; N is FK particle count; K is averaging samples. B20 counts batched APIs, not native episodes or physics controls. Source rows and native policy rows are also distinct.

QGF 27 kappa levels span0..320; M {8,16,32,64}. Literal level lists are in LITERAL_GRID_LEVELS_FROM_FULL_SAVED_TABLES.csv. Full historical QGF 648 / FK 2160 / DF 27720 tables remain referenced unchanged, with SHA and row counts. Their staged designs are not one exhaustive Cartesian claim.

Evidence, resources and remaining work

This is a local, reviewable partial report. No ZIP, upload, new model or numerical run was performed to make it.

Four final fullstate references - no checkpoint payloads embedded

Goal / seed / phase	Manifest SHA prefix	Payload SHA prefix
3 / 11 / BC	df520049d467e3f3	247783361b8888a6
3 / 11 / QV	de0c33a88fa1b683	33bf6e496d84f684
3 / 23 / BC	43d3affd995908fb	87d0d85f537da245
3 / 23 / QV	e4b347107eb430dd	b7f86ce78f7f947e

The checkpoint reference JSON contains complete paths, exact manifest/payload SHA256, declared sizes and audit links. Only manifest JSON was SHA checked for this report; payload bytes were not opened, rehashed, copied or decoded. No weights, raw arrays or portable runtime are embedded.

Same 24-hour scientific budget, frozen at plan 496

Accounting item	Seconds
Measured outer-parent actual, each once	50,837.502716505434
Separate old UNKNOWN conservative charge	600.000000000000
Used / committed at cutoff	51,437.502716505434
Remaining of 86,400	34,962.497283494566
Active scientific caps at cutoff	0

Worker/watchdog/helper/admin clocks are not added again. Earlier timings are context, not cold-cost, RSS, output-fit, speedup or completion bounds. Fourteen 50k phases, validations, fresh CONF and all 33 primary gates remain unfinished.

Delivery coverage and next evidence

The public plan 409 archive freezes Nov 8 23:35:08 UTC: 241,048 eligible files /49.138GB /38 distinct PDFs, with installed-runtime/cache/git/OS exclusions. Later Quad work is outside that snapshot. Public P1 code+QA excludes raw arrays and later weights. Old plan 457 PDF remains immutable; this report is not uploaded.

Next: remaining training, family-specific reviewed engineering/readiness, official fresh-reset native VAL and fresh CONF. No hidden full-goal BC cutoff, seed/cohort cut, forced winner or KEEP_ALL33 waiver.

Browse existing reports: github.com/diwuz1-dev/dflow-qgf-research-handoff/tree/main/reports Archive: releases/tag/complete-campaigns-20261108T233045Z Exact local reference links,18 frozen-source hashes, CSV provenance and full table pointers are in REPORT_SOURCE_AND_DELIVERY_REFERENCES.json.